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TITLE:StoryLab Voice
A Voice-Guided Educational Ecosystem with AI Assistant, Multi-Subject Stories, and Virtual Labs

System Modeling: Use Case Analysis

1. Actors (Users)

- **Student:** Primary rural learner who uses voice/touch to interact with stories and virtual labs.
- **Teacher:** Supervisor who assigns modules, monitors student progress, and views reports.
- **Administrator:** System engineer who manages user roles, AI voice models, and system configurations.
- **AI Voice Assistant:** Central intelligent subsystem coordinating speech recognition, TTS, and core app engines.

2. Primary Use Cases

- **UC-01: Interact via Voice Chatbot:** Bidirectional spoken tutoring, Q&A, and navigation.
- **UC-02: Learn via Multi-Subject Stories (StoryLearn):** Interactive audio story modules with embedded questions.
- **UC-03: Perform Virtual Science Experiment (LabInBox):** Voice-guided 2D virtual lab execution.

- **UC-04: Process Offline Speech:** Local Speech-to-Text (STT) and Text-to-Speech (TTS) conversion.
- **UC-05: Sync Offline Data:** Local data storage and periodic cloud synchronization.
- **UC-06: Assign Modules & Monitor Progress:** Teacher curriculum assignment and analytics tracking.
- **UC-07: Manage System & AI Configuration:** Admin audit logs and AI voice model deployment.

3. Key Relationships & Dependencies

Include Relationships (`<<include>>`)

- **UC-01 (Voice Chatbot)** `<<include>>` **Convert Speech to Text** *(Always parses student audio input)*
- **UC-01 (Voice Chatbot)** `<<include>>` **Convert Text to Speech** *(Always generates spoken AI response)*
- **UC-02 (StoryLearn)** `<<include>>` **UC-01 (Voice Chatbot)** *(Stories are narrated via chatbot)*
- **UC-03 (LabInBox)** `<<include>>` **UC-01 (Voice Chatbot)** *(Virtual labs are directed via chatbot)*
- **UC-02 / UC-03** `<<include>>` **Store Data Locally** *(Saves progress offline automatically)*

Extend Relationships (`<<extend>>`)

- **Provide Spoken Hint** `<<extend>>` **UC-01** *(Triggers conditionally when student is stuck)*
- **Generate Lab Report** `<<extend>>` **UC-03** *(Executes upon experiment completion)*
- **Sync Local Data to Cloud** `<<extend>>` **UC-05** *(Triggers when internet connection is restored)*

Generalization (Inheritance)

- **Student** `is-a` **User** *(Inherits `userId`, `name`, `language`; adds `age`, `learningLevel`)*
- **Teacher** `is-a` **User** *(Inherits `userId`, `name`, `language`; adds `schoolId`, assignment tools)*
- **Administrator** `is-a` **User** *(Inherits `userId`, `name`, `language`; adds `adminId`, management rights)*

USE-CASE DIAGRAM:

